

CMSC 128: Introduction to Software Engineering

Software Architecture Document

Barcode-based Museo Kordilyera Collection Information System

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June 15, 2023

Revision History

Version	Description of Versions / Changes	Date
1.0	Prototype 1 - Initialization of database and tables - Initial user interface for pages - Basic login system - Inventory using accession number of materials - Table pagination	
2.0	Prototype 2 - Improvement of existing user interface - Improvement of database structure - Material search system - Add error handling for existing forms - Add function has been added	
3.0	Prototype 3 - Improvement of existing user interface - Output created tables into Excel file for printing - Implement admin privileges for some staff - Implement searching for created tables - Data backup and restore system - Edit function has been added - Finalize the codes for the existing features	
4.0	Final Prototype - Remove restore function due to risk - Improved User Interface	

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Software Architecture Document

1. Introduction

This paper gives an in-depth description of the Barcode-based Museum Collection Information System and describes its design.

The document outlines the architecture's objectives, the use cases that the website supports, architectural styles, and the functionalities that have been requested. From the initial conceptualization through the final outcome, the document explains the reasoning behind the architectural and design choices.

1.1. Purpose

The Software Architecture Document (SAD) provides a comprehensive architectural overview of the Barcode-based Museum Collection Information System. It presents several different architectural views to depict the different aspects of the system.

In this paper, the use case and deployment view models are used to discuss the system architecture from two separate points of view. This paper describes the system's static and dynamic behavior and contains all the necessary diagrams and descriptions.

It is feasible to create a system from the end user's point of view by utilizing the use case view model. It demonstrates specifically how various user types interact with the system to address a challenge. On the other hand, using the Deployment view model demonstrates how processing is distributed over a number of system nodes, including how processes and threads are physically distributed.

1.2. Scope

The scope of this SAD is to explain the architecture of the Barcode-based Museum Collection Information System.

This document describes the various aspects of the Barcode-based Museum Collection Information System website design that are considered to be architecturally significant. These elements and behaviors are fundamental for guiding the construction of the website and for understanding this project. Stakeholders who require a technical understanding of the Barcode-based Museum Collection Information System are encouraged to start by reading the Project Proposal, and this document [PP, SAD].

1.3. Definitions, Acronyms, and Abbreviations

- **Admin** – Administrator of the application's database
- **Ajax** – Asynchronous JavaScript and XML
- **Apache** – Web Server
- **Bootstrap** – Open-Source CSS Framework
- **CSS** – Cascading Style Sheets
- **Datatables** – plugin for the jQuery Javascript Library
- **HTML** – Hypertext Markup Language
- **Javascript** – Programming Language
- **jQuery** – Javascript Library
- **MySQL** – Database Service
- **PHP** – Hypertext Preprocessor
- **SAD** - Software Architecture Document
- **SQL** - Programming language used for database management
- **TXT** – Computer file that contains only text and has no special formatting
- **User** - The person/s who will use the the application (admin/guest)
- **XAMPP** – cross-platform, Apache, MySQL, PHP and PERL

1.4. References

[PP]: Project Proposal

[SAD]: System Architecture Document

[Mooney, Shimpi, Osman, and Pendergrass]: Sample SAD,

https://projects.cecs.pdx.edu/attachments/download/3146/Software_Architecture_Document.docx

[Patel]: Dynamic and Static Behavior of the System,
<https://eanswered.blogspot.com/2012/07/dynamic-and-static-behaviour-of-system.html>

[SCE]: Concepts: Deployment

View, <https://sceweb.uhcl.edu/helm/RationalUnifiedProcess/process/workflow/analysis/codview.htm>

[VisualParadigm]: What is Activity Diagram?,
<https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-activity-diagram/>

[VisualParadigm]: What is use case Diagram?,
<https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-use-case-diagram/>

[UTM]: Concept: use case Model,
https://www.utm.mx/~caff/doc/OpenUPWeb/openup/guidances/concepts/use_case_model_CD178AF9.html

2. Architectural Goals and Constraints

The following are some key requirements and system constraints that have a significant bearing on the architecture:

1. Through XAMPP-launched local Apache and MySQL servers connected to a host network, the web application will be accessed. The minimal resources available were taken into account when designing this process.
2. The system is needed for more than nine months. This is because the optimized database management provided by the system will be useful for the client because of its faster, more reliable, and more accurate mechanisms in managing loads of data.
3. The system will have a lifetime of more than four years because it runs up to date-alongside many database management systems.
4. Every piece of artifact will have a special accession number based on the format provided, as well as a special barcode that will be used for effective automated inventory management. For users and employees handling artifacts data, the system will also grant administrator rights.
5. The system architecture's main objective is to reduce client workload and boost overall performance. This is accomplished by developing automated methods for managing the inventory of museum artifacts and the ability to produce the required reports of information.
6. The system will have a user-friendly interface that will make automated information processing functionalities that are specially designed for the museum inventory management process readily available to staff and guest users.

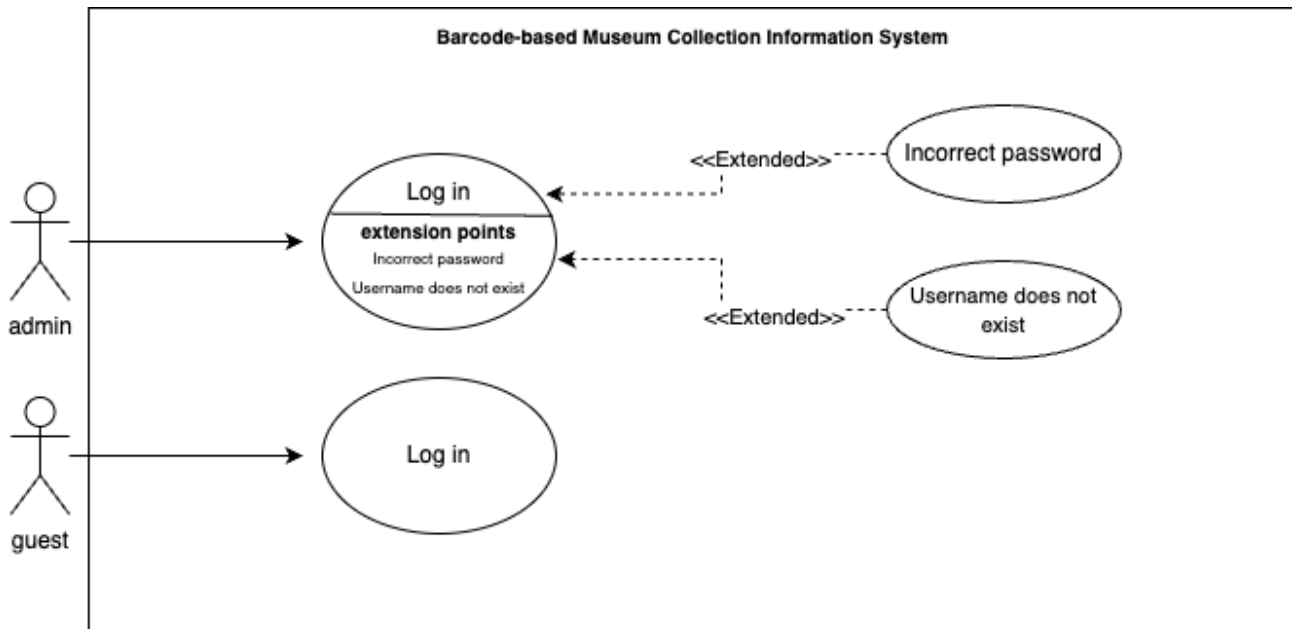
3. Use Case View

A frequent modeling tool for a new software program being built with the goal of describing the expected operation of the software is a use case view or diagram. Use case diagrams can help analysts and domain specialists establish a system's context, capture a system's requirements, validate a system's architecture, drive implementation and produce test cases.

3.1. Use Case View

The use case diagrams that show every aspect of the created software are listed and discussed in this section. The paper also covers the sequence of events that corresponds to the systems' intended behavior.

3.1.1. Login Use Case View



3.1.1.1. Name and Description of the Use Case View

This use case diagram shows the use cases involved in the login system. This describes the processes involved in logging into the system, along with other scenarios that prevent user login.

3.1.1.2. Flow of Events of the Use Case View

The main flow of this use case diagram begins when the user visits the page and is not logged in. The web portal will show a page that requests the username and password of the user. After trying to log in, there are 2 possible scenarios that prevent the user login, namely (1) incorrect password and (2) username does not exist, which are considered subflows and will be considered individually.

1. Incorrect Password

This subflow begins when the user tries to login with a correct username but with a wrong password. The web portal will display an error that reads 'The password is incorrect' above the password field of the login form.

2. Username Does Not Exist

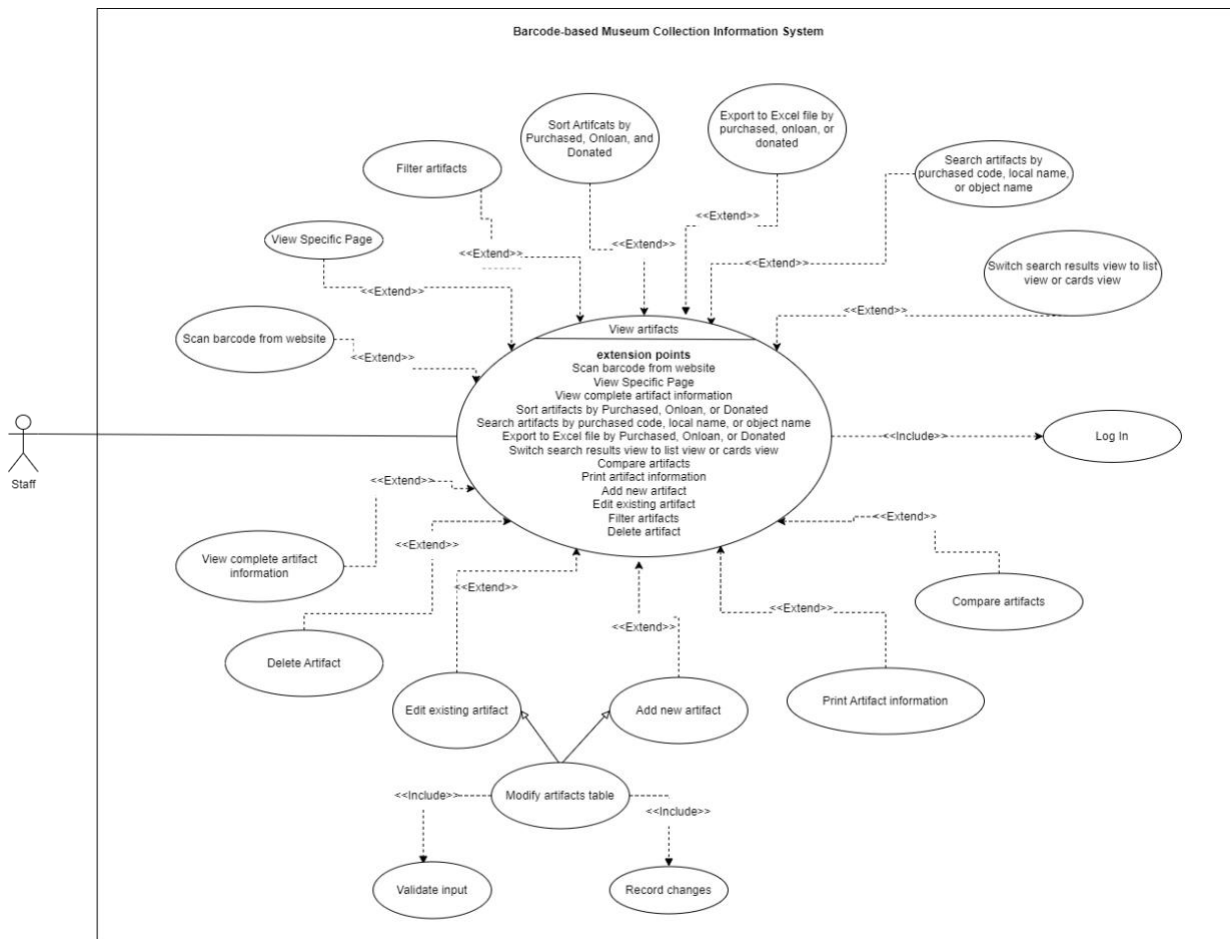
This subflow begins when the user tries to login with an incorrect username.

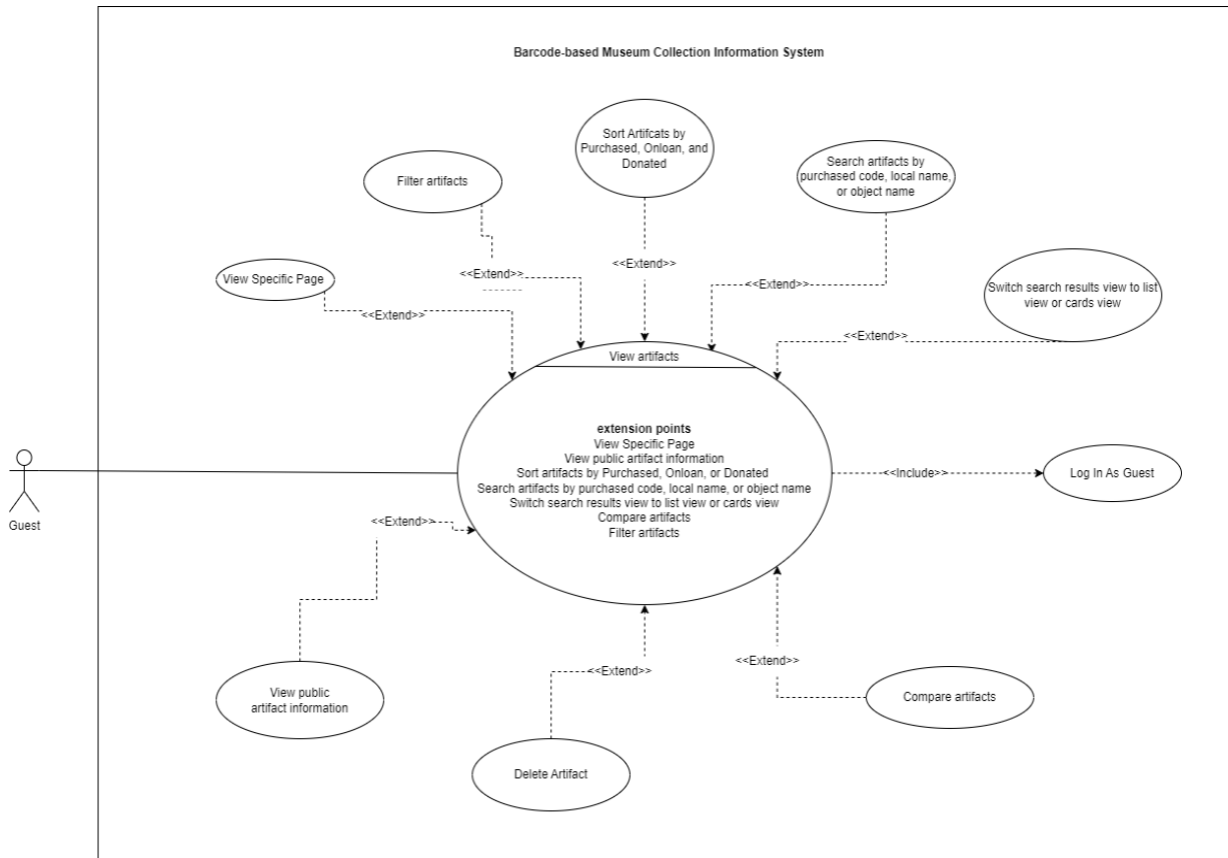
The web portal will display an error that reads ‘The username does not exist’ above the username field of the login form.

3.1.1.3. Relationships Among Use Cases

This part discusses some relationships that exist between some use cases. As shown in the diagram, and is also evident in the flow of events, the main use case for this activity is the login system. The several use cases, namely incorrect password, username does not exist, and account has been deactivated extend the login use case, allowing the various scenarios to be more explicitly considered.

3.1.2. Artifacts Management Use Case View





3.1.2.1. Name and Description of the Use Case View

The following use case diagrams show the use cases involved in the management of artifacts both by the staff and guest user. This describes the processes involved mainly in viewing, searching, filtering, sorting, adding, editing, and comparing the artifacts stored in the database, along with other features depending on the degree of privilege by the user.

3.1.2.2. Name and Description of the Use Case View

We consider two flows of the use case diagrams.

Admin Use Case View. The flow of the use case diagram for the staff begins when the staff logs in to the system and proceeds into the landing page for staff/administrator. The landing page will show the clickable cards listing all the artifacts stored in the database, sorted by Purchased first. It will also display other options in the page namely (1) searching by catalogue number, local name, or object name, (2) scanning a barcode, (3) filtering search results, (4) sorting by other categories, (5) changing display of results, (6) exporting the database by category, (6) clickable cards/links, (7) comparing

artifacts, (8) adding new artifacts. These subflows will be discussed individually.

1. Searching By Catalogue Number, Local Name, or Object Name

This subflow begins when the staff inputs to the search bar by catalogue number, local name, or object name or no input at all, and clicks enter by the keyboard or the search button located at the right side of the search bar. The displayed cards will change according to the user's input upon pressing enter or the search button. If the input is empty, then the entire database by the sorted category (Purchased, Onloan, or Donated) will be returned. If the search input is not present in that category in the database, then no cards will be displayed but instead a text informing the staff that no results have been found.

2. Scanning a barcode

This subflow begins when the user clicks the 'Scan' button found at the left side of the search bar. When pressed, a pop-up window will appear and prompt the user for camera permission to start scanning. After allowing access, the scanner will start, and the user shall place a barcode directly at the scanner. After recognizing the barcode, the scanner will automatically search for the corresponding item in the database depending on the current category and display the result.

3. Filtering Search Results

This filter search option is found at the first column below the search button. This subflow begins when the staff ticks any of the checkboxes and presses the filter search button (found at the bottom of the filter options). Upon filtering, the displayed search results will be updated according to the filter request by the user.

4. Sorting By Other Categories

This subflow begins when the staff clicks a category from the three primary categories located at the upper-leftmost second column below the search bar, right beside the filter search column. There are three categories: Purchased, Onloan, and Donated.

5. Changing Display of Results

This subflow begins upon clicking on the button with a 'List' icon found at the opposite end of the sort by categories. When clicked, the cards displayed will be converted into a table form of lists of search results. Accordingly, the user can toggle back into the 'Card' display view by clicking the button with a picture icon found at the same location as the

list button. The user can toggle between these displays back and forth.

6. Exporting The Database by Category

This subflow begins when the user clicks on the ‘Export’ button found beside the list view/card view button. Once clicked, an excel file will be downloaded containing the entire database information of the current category.

7. Clickable Cards/Links

This subflow begins when the user clicks on the cards/links displayed from the search results according to the request of the user. Upon clicking, the user will be redirected to a page containing the detailed information of the specific artifact. Here, there are three primary sub-subflows that will be considered: print, edit, and delete artifact information.

7.1. Print Artifact Information

This sub-subflow is a ‘Print’ button that once clicked, will prompt a print window with the artifact information and the museum’s customized worksheet format.

7.2. Edit Artifact Information

This sub-subflow is an ‘Edit’ button that once clicked, will redirect the user to a new page that will make the entries editable. The user can then either cancel or save the changes on the artifact.

7.1. Delete Artifact Information

This sub-subflow is a ‘Delete’ button that once clicked, will prompt a window that will ask the user to confirm that this item will be deleted permanently. Once clicked, that certain artifact along with its information will be deleted from the database.

8. Comparing Artifacts

This subflow begins when the user clicks on the ‘Compare’ option located in the navigation bar of the webpage. In the compare page, the user can input the catalogue numbers of the artifacts he/she wishes to compare starting with two columns (one for each artifact information). Additionally, the user can add up to four columns to compare artifacts and remove the additional two columns as he/she wishes to.

9. Adding New Artifacts

This subflow begins when the user clicks on the ‘Add’ button on the navigation bar. The ‘Add’ page will display a list of input fields according to the three main categories (Purchased, On Loan, Donated) that requests input regarding the information of the new artifact, and when the user clicks on ‘Add, the webpage will display either a confirmation message and the corresponding barcode of the new artifact if all inputs are valid, or some error messages if some inputs are invalid.

Guest Case Use View. The flow of the use case diagram for the guest begins when the guest logs in to the system with the ‘Log-in as Guest’ button and proceeds into the landing page for guest users. The landing page will show the clickable cards listing all the artifacts stored in the database, sorted by Purchased first. It will also display other options like the staff/administrator but with limited access/privileges. The only allowable options for the guest relative to the options discussed above in the staff privileges are: (1) searching by catalogue number, local name, or object name, (3) filtering search results, (4) sorting by other categories, (5) changing display of results, (6) clickable cards/links, and (7) comparing artifacts.

The guest will also have limited information about (6) clickable cards/links, wherein the guest will not be able to access the ‘Print’, ‘Edit’, and ‘Delete’ functions unlike the staff. Moreover, the information listed on the page displayed on (6) or the detailed information of each artifact will be limited to non-confidential or public information only. This will also apply to (7), when the guest user chooses to compare artifacts.

3.1.2.3. Relationships Among Use Cases

This part discusses some relationships that exist between some use cases. As shown in the diagram, and is also evident in the flow of events, the main use case for this activity is viewing artifacts relative to the privilege of the user. Additional use cases which extend the viewing of material are viewing specific pages when clicking on cards/links of the search results, filtering, sorting, changing display view, exporting, scanning of barcodes, comparing and searching artifacts, adding new artifacts, generating barcodes, editing existing artifacts, and deleting artifacts.

Furthermore, the login use case is included in viewing the materials since the user must log in to the page first to see the database's contents.

3.1.2.4. Related Use Cases

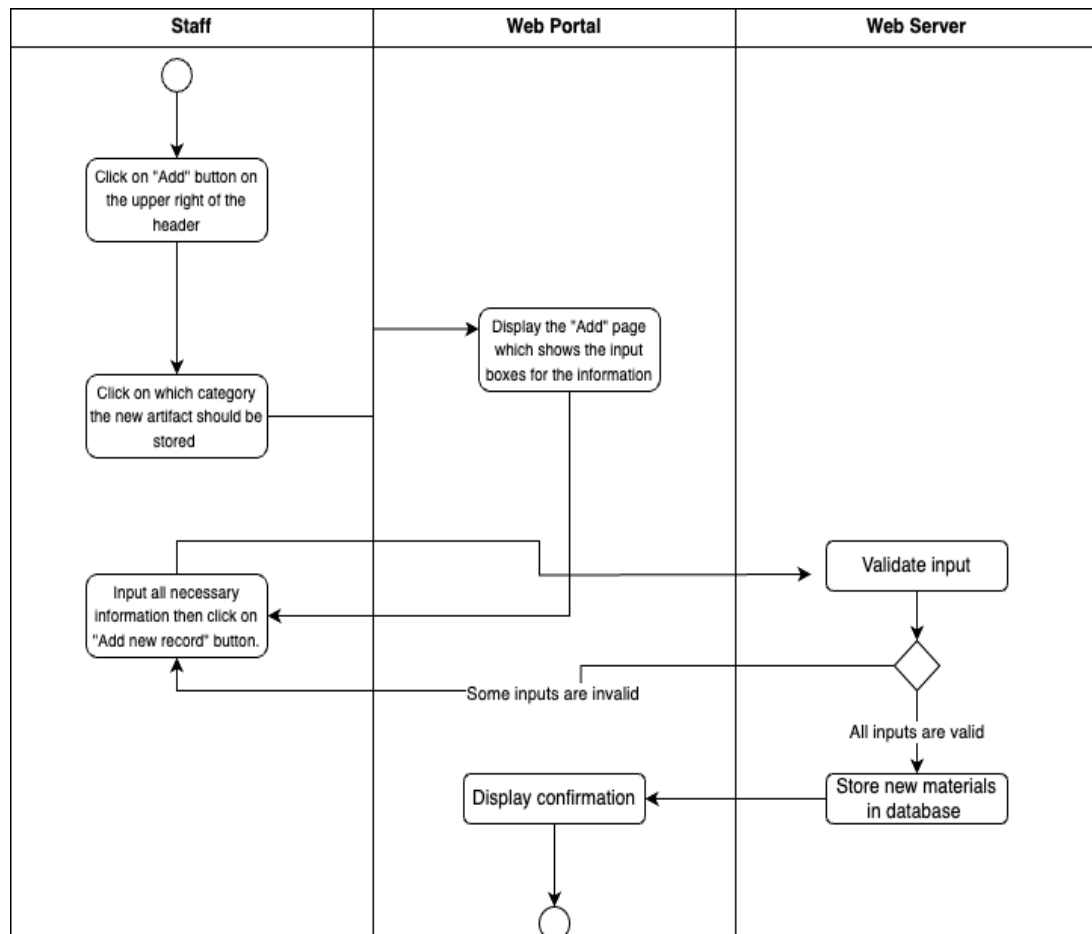
Since signing into the system for authentication is necessary before accessing all the items stored in the database, the login use case is included in this particular use case diagram.

3.2. Activity Diagram

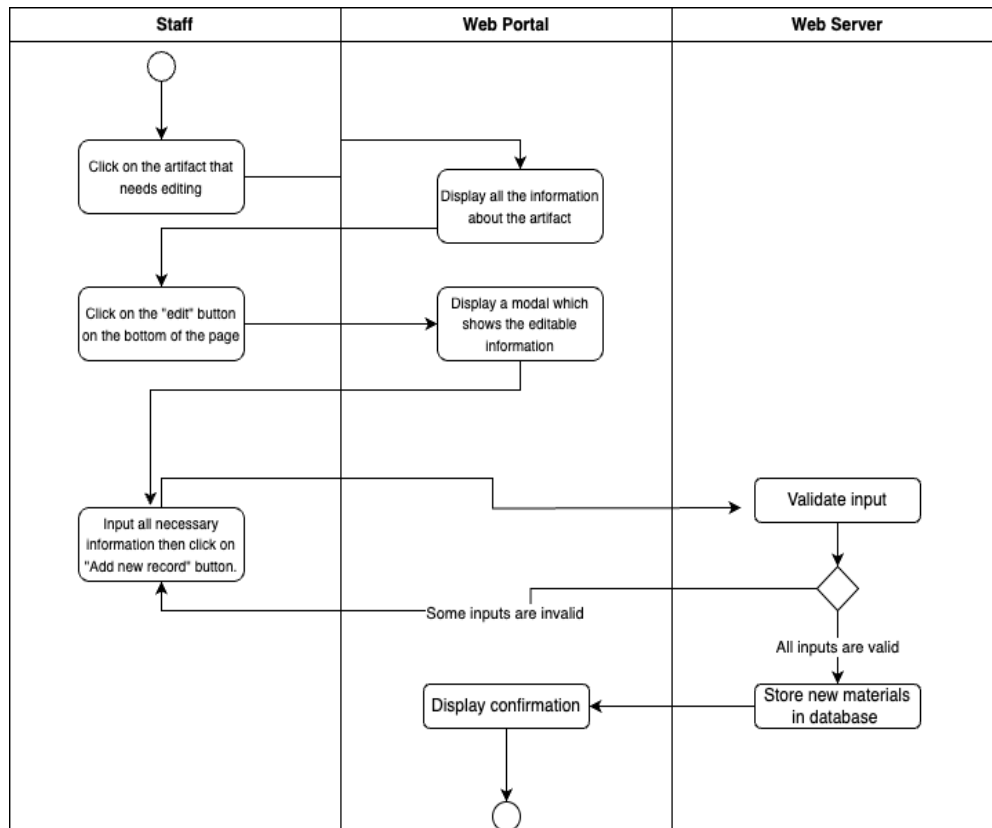
An activity diagram is an important behavioral diagram that describes dynamic aspects of the system. This part shows the activities that can be done given the various functionalities of the software. The diagrams make use of a swim lane in order to demonstrate the interaction mainly between the system and the users of the software.

3.2.1. Modifying Artifacts Table

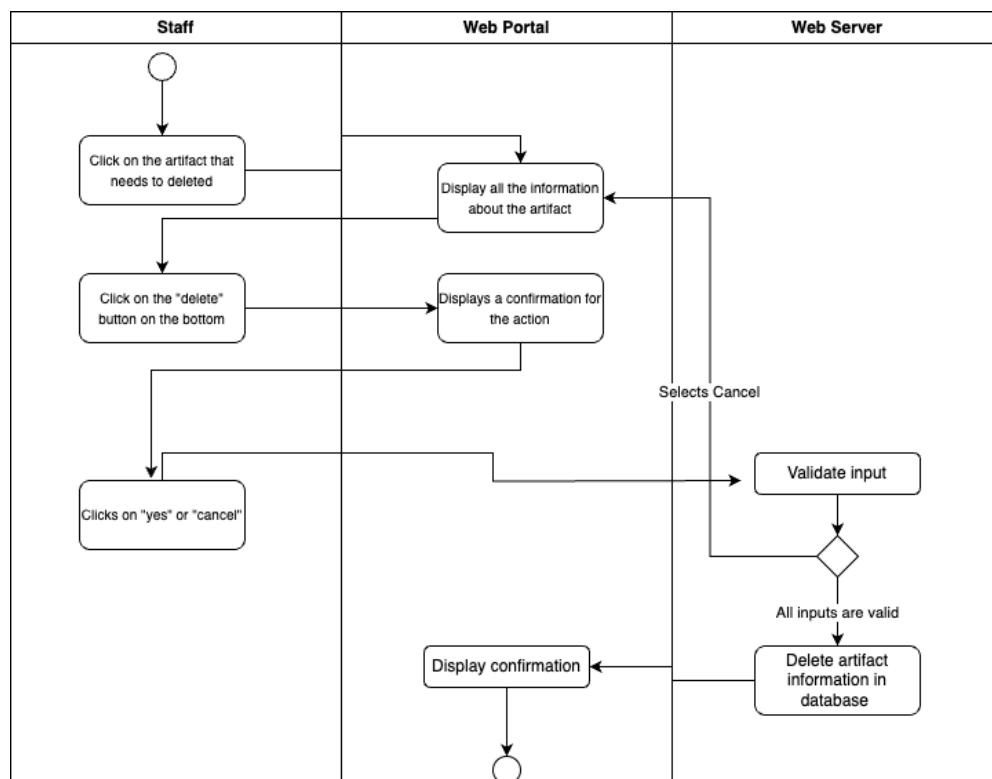
3.2.1.1. Adding New Artifact



3.2.1.2. Edit Existing Artifact

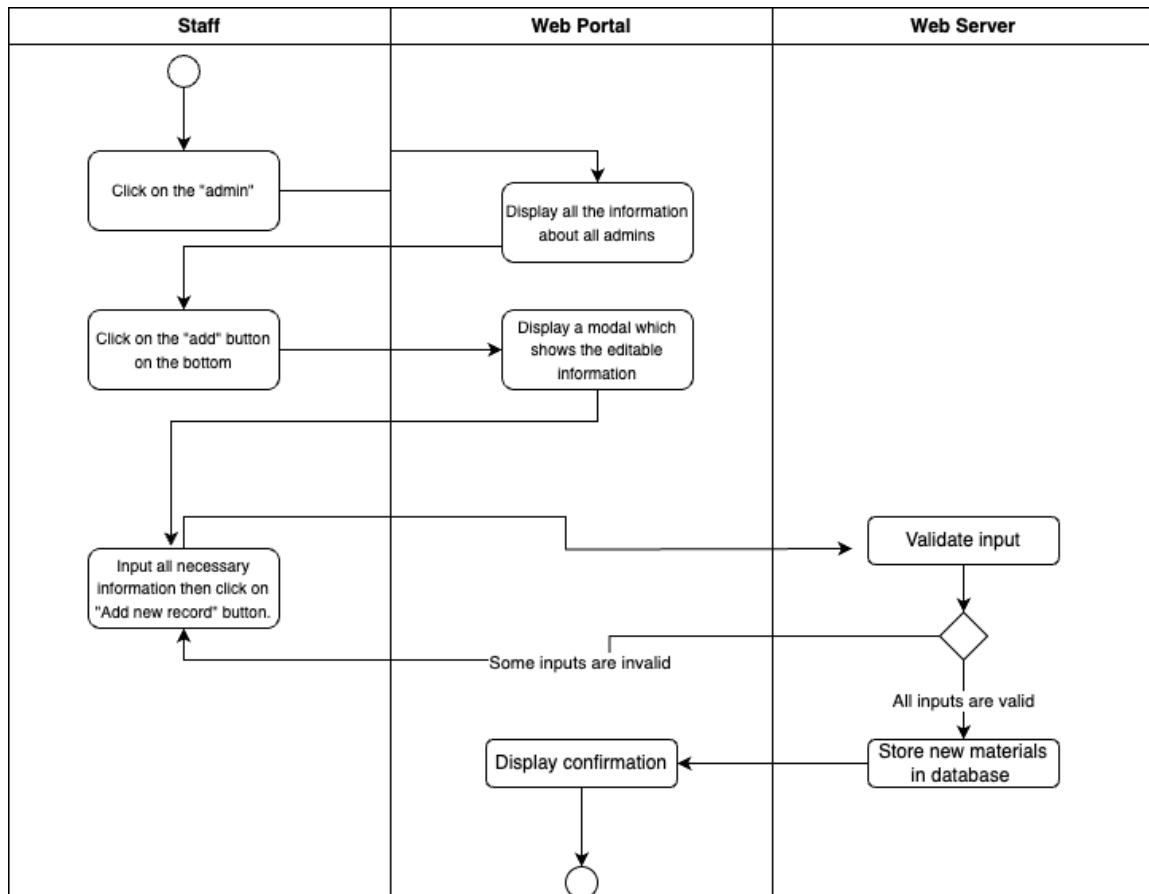


3.2.1.3 Delete Existing Artifact

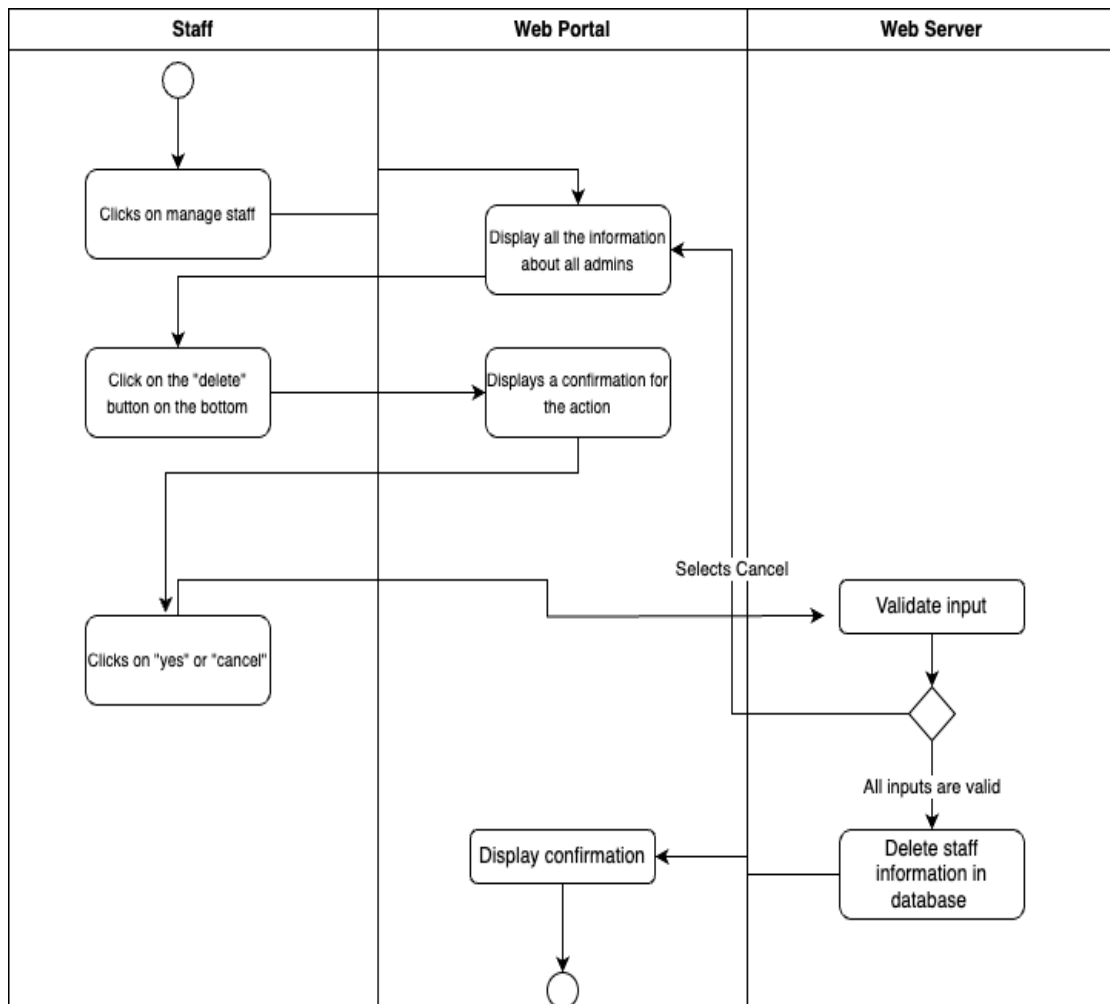


3.2.1. Staff Management

3.2.1.1. Add New Staff



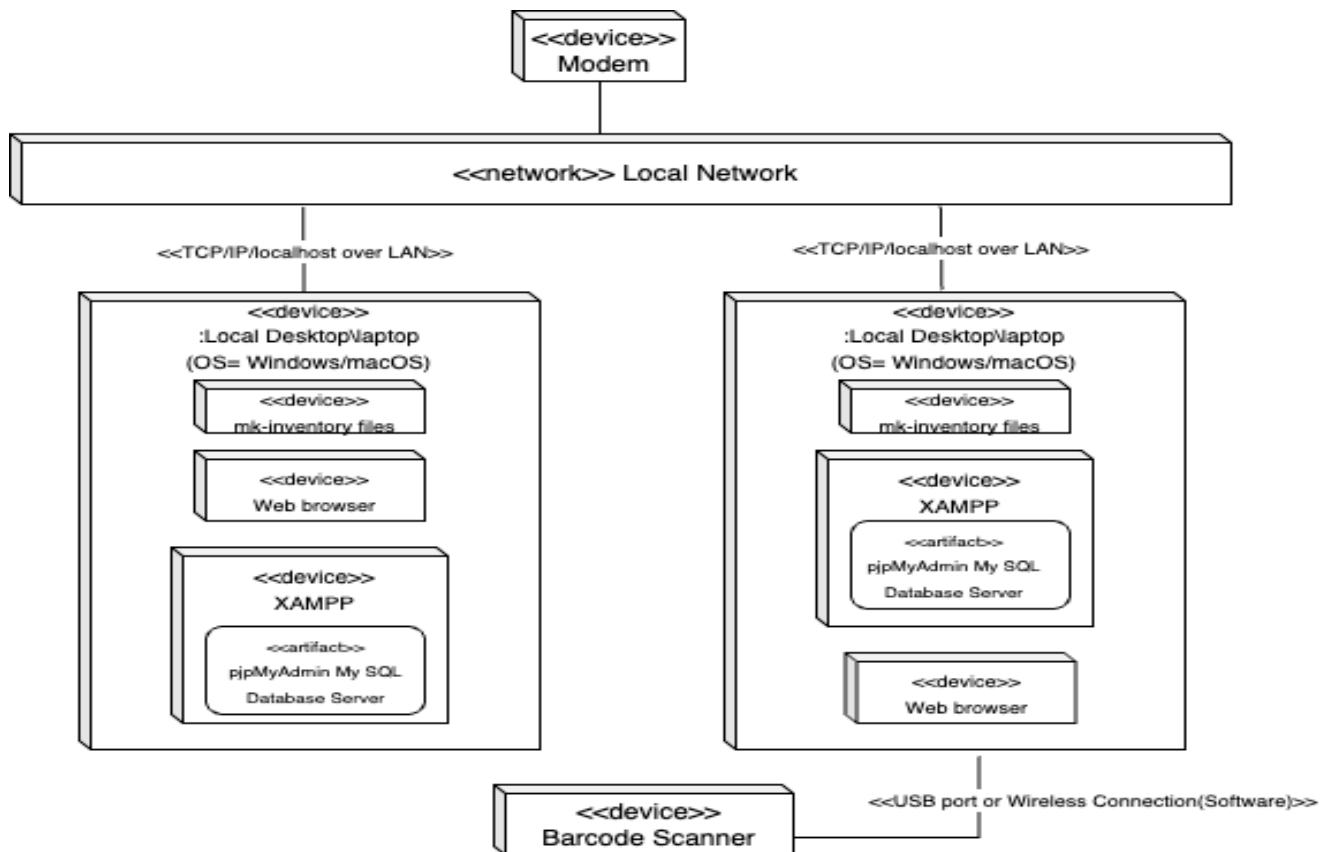
3.2.1.2. Delete Staff



4. Deployment View

An XAMPP software and the web application files will be installed on a single host PC/network. The host network shall launch a local server of Apache and MySQL using the XAMPP platform. Multiple user PC/devices/nodes may access the web application through TCP/IP/localhost over the local area network. Each user device may have a barcode scanner connected via USB port or wireless connection.

The application's deployment specifics can be seen below:



5. Size and Performance

The chosen software architecture supports the key sizing and timing requirements:

1. Up to 12 users may concurrently access the local servers using the system at any given time.
2. Within 2-3 mins, 80 percent of the transactions in the system should be done.
3. The Barcode-based Museum Collection Information System website shall require around ~1.40 disk space (for XAMPP installation and UPB-LIS program files) and 450MB RAM (using Google Chrome) for program execution.

The selected architecture supports the sizing and timing requirements through the implementation of a client-server architecture. The client portion is implemented on local campus PCs or remote dial up PCs. The components have been designed to ensure that minimal disk and memory requirements are needed on the PC client portion.

6. Quality

The software architecture supports the quality requirements:

- The desktop user-interface shall be Windows OS and Mac OS compliant.
- The user interface of the Barcode-based Museum Collection Information System shall be designed for ease-of-use and shall be appropriate for a computer-literate user community with no additional training on the System.
- The software shall only be accessible during the operating hours of Museo Kordilyera from Monday to Friday, 9:00 AM to 4:00 PM.

Appendices

Appendix A - CRUD Functionalities

A.1. Search and Display Artifact Information

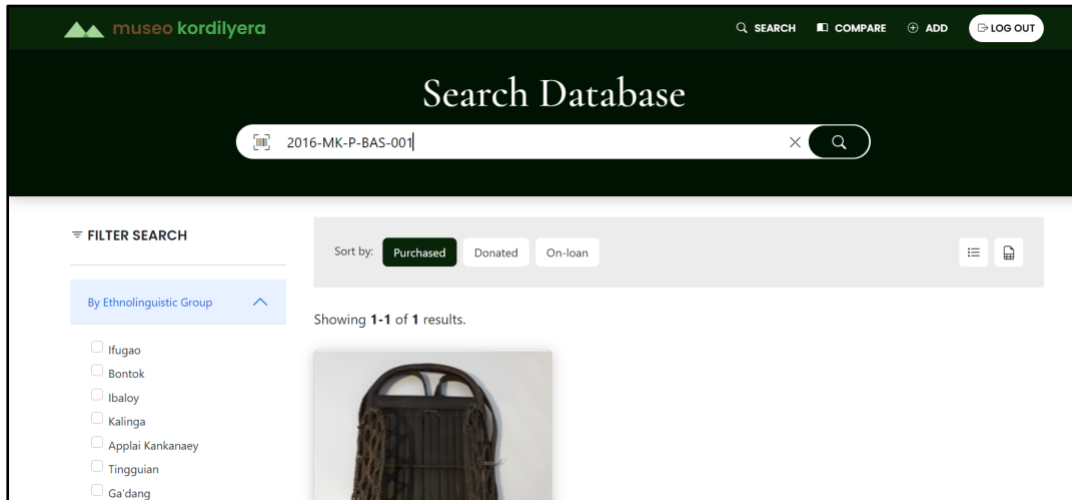


Figure 1: Searching for a Specific Artifact

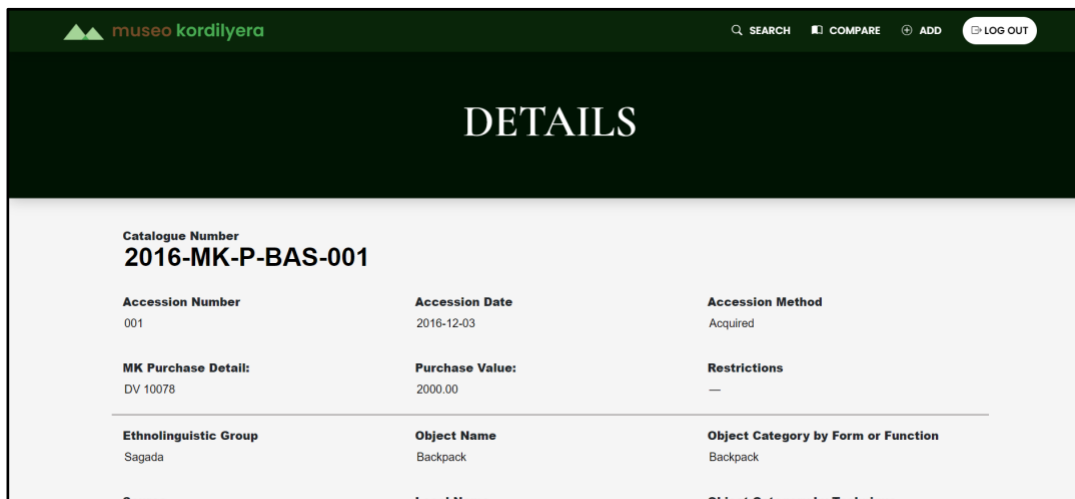


Figure 2: Various Information Displayed for an Artifact

Search by selecting the search bar at the top of the page and typing the code/local name/object for the artifact you wish to search, then clicking on the search icon. Once the artifact/s have been searched and filtered, click on the artifact card to view the various information as shown in Figure 2.

A.2. Deleting an Artifact

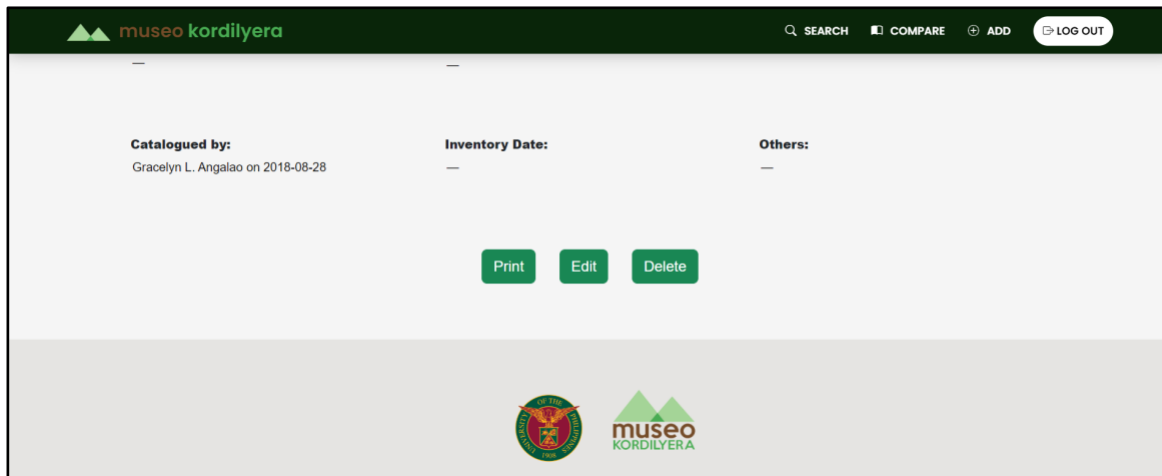


Figure 3: Button group found at the bottom of artifact information page

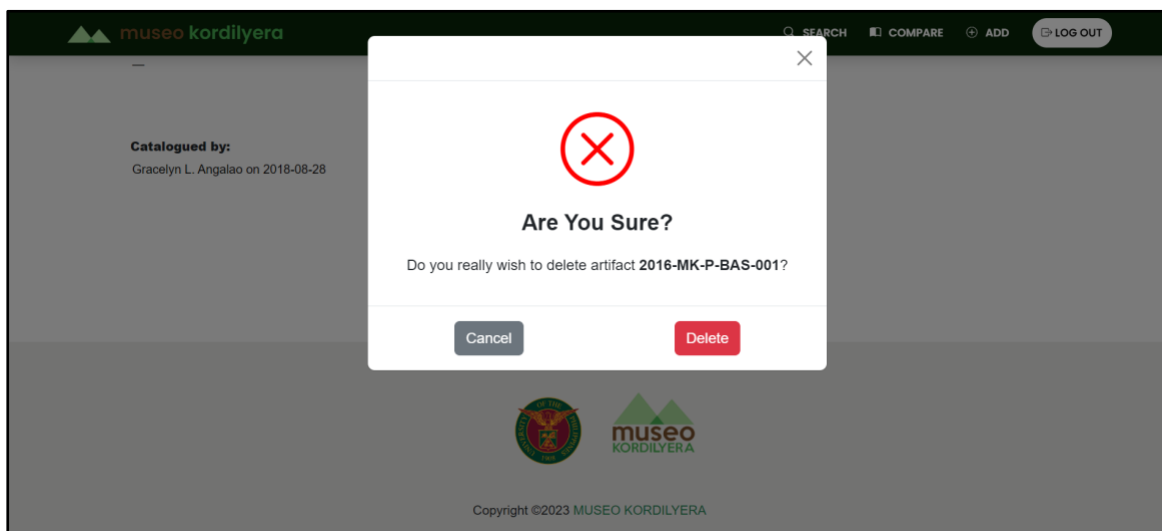


Figure 4: Alert Box after clicking on Delete

On the information page about an artifact, scroll to the bottom of the page to locate the group button for various actions. To delete the current artifact, click on the delete button and a pop-up will appear for confirmation. Clicking the delete button on the pop-up will commit the action and delete the artifact from the database.

A.3. Editing an Artifact's Information

The screenshot shows the 'Edit Artifact' page for '2016-MK-P-BAS-001'. The form is divided into several sections: 'Accession Number' (001), 'Accession Date' (2016-12-03), 'Accession Method' (Acquired), 'MK Purchase Detail' (DV 10078), 'Purchase Value' (2000.00), 'Restrictions' (—), 'Ethnolinguistic Group' (S), 'Object Name' (Backpack), 'Object Category by Form or Function' (Backpack), 'Local Name' (Chagee), and 'Object Category by Technique' (Basketry). A dropdown menu for 'Ethnolinguistic Group' is open, showing options: 'Isinay' (selected), 'Isnag', 'Sagada', 'Ga'dang (Paracelis)', 'Sample', and 'Sample'. At the bottom, there is a 'Location History' table with columns 'Start' and 'End'.

Figure 5: Fields become editable after clicking edit button

To edit an artifact's data, click the edit button found at the bottom of the information page (displayed in Figure 3). After clicking on the button, the fields that were originally readonly can now be updated. Some fields have autocomplete on them since they have choices available. Nevertheless, new values on these fields will be automatically created and added to the choices for future updates.

A.4. Adding an Artifact

The screenshot shows the 'Add to Database' form. At the top, there is a header with the Museo Kordilyera logo and navigation links: SEARCH, COMPARE, ADD, and LOG OUT. Below the header, the title 'Add to Database' is displayed. Underneath, there is a 'Sort by:' section with three buttons: 'Purchased' (selected), 'Donated', and 'On-loan'. The form fields are arranged in a grid: 'Accession Number:' (text input), 'Accession Year:' (text input with '2023'), 'Category e.g(BAS):' (text input), 'Accession Date:' (text input with 'dd/mm/yyyy' and a calendar icon), 'Accession Method:' (text input), 'Restrictions:' (text input), 'MK Purchase Detail:' (text input), and 'Purchase Value:' (text input).

Figure 6: Adding a new Artifact to the Database

To create and add a new artifact entry into the database, first click on the add button located at the header (the header is found at the top of the page). Select the type of artifact based on acquisition: purchased, donated, or on-loan. The following steps are only a matter of completing the necessary fields. After clicking the “save the record” button found at the bottom of the page, the artifact is now added to the database.

Appendix B - Helper Functionalities

B.1. Compare Artifacts

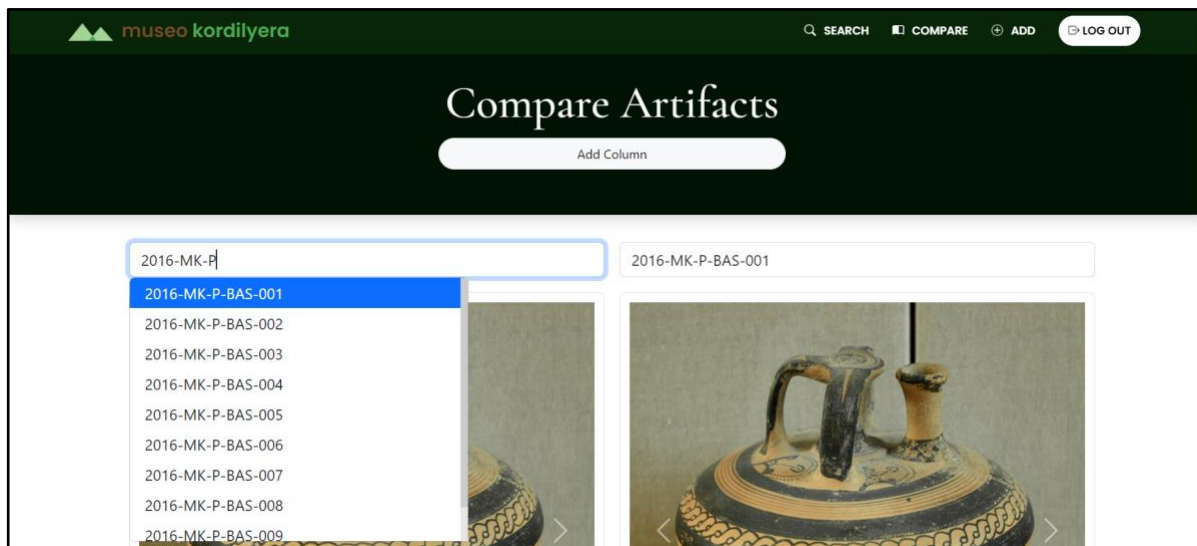


Figure 7: Default compare configuration

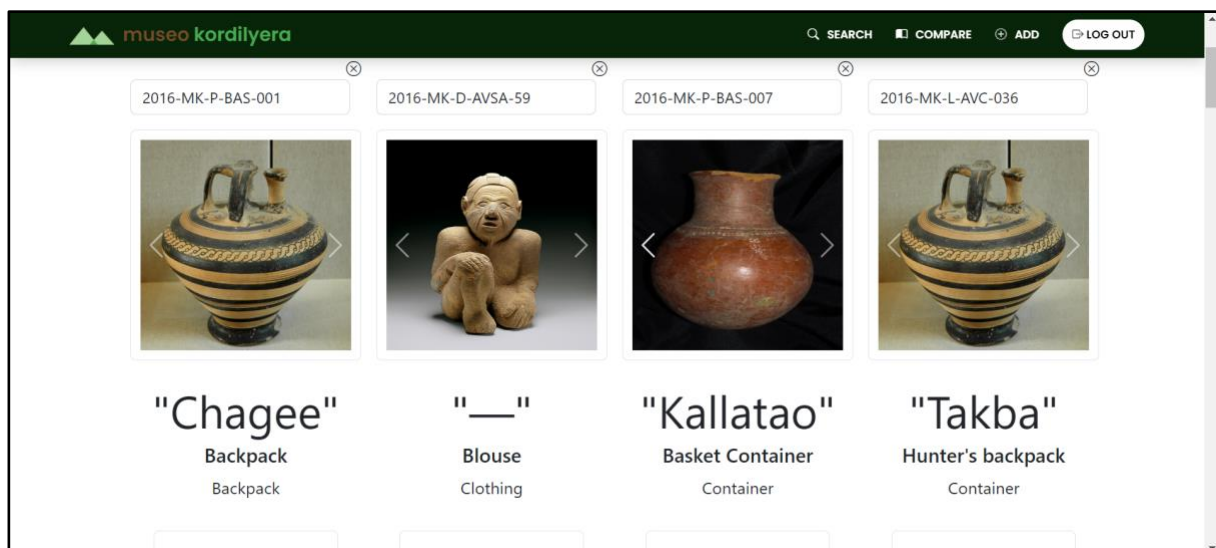


Figure 8: Example custom artifact compare configuration

The default configuration for comparing artifacts is 2 columns, both being the artifact with code 2016-MK-P-BAS-001. The compare input bar only accepts artifact codes unlike the search bar since some artifacts have similar or unavailable local name/object names. Columns can be added up to a maximum of 4 and configuration is stored in cookies so that even if switching pages, the configuration is persistent.

B.2. Filter Search Results

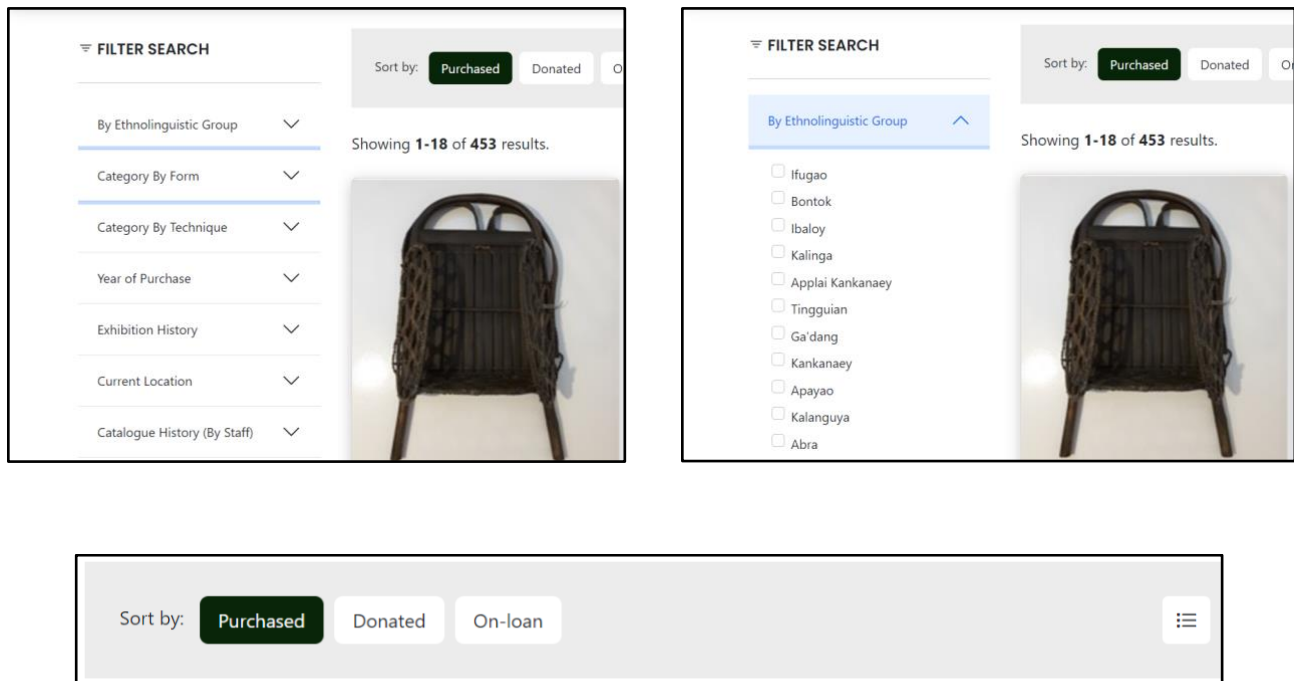
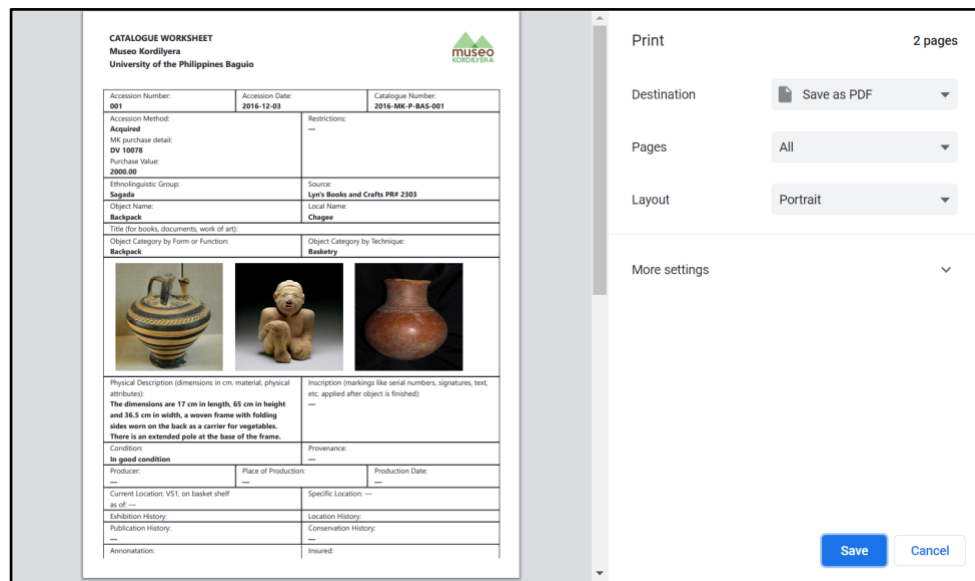


Figure 9: Various filter options

The search results can be filtered on the Filter Search sider on the search page. Select all options you wish to filter by and click the button at the bottom to filter the results. Additionally, the results can be sorted according to the three main categories: purchased, donated and on-loan. The results can also be displayed in grid or list view depending on what the user prefers.

Appendix C - Export/Import Functionalities

C.1. Print Catalogue Worksheet



CATALOGUE WORKSHEET		Museo Kordilyera University of the Philippines Baguio	
Accession Number: 001	Accession Date: 2016-12-03	Catalogue Number: 2016-MM-P-BAS-001	
Accession Method: Acquired	Restrictions: ---		
MR purchase detail: DV 10078			
Purchase Value: 2000.00			
Enthnographic Group: Segada	Source: Lyn's Books and Crafts PRP 2383		
Object Name: Backpack	Local Name: Chagne		
Title for books, documents, work of art:			
Object Category by Form or Function: Backpack	Object Category by Technique: Basketry		
			
			
Physical Description (dimensions in cm, material, physical attributes): The dimensions are 17 cm in length, 65 cm in height and 34.3 cm in width, a woven frame with folding sides worn on the back as a carrier for vegetables. There is an extended pole at the base of the frame.		Inscription (markings like serial numbers, signatures, text, etc. applied after object is finished): ---	
Conditions: In good condition		Provenance: ---	
Producer: ---	Place of Production: ---	Production Date: ---	
Current Location: V51, on basket shelf		Specific Location: ---	
Exhibition History: ---		Location History: ---	
Publication History: ---		Conservation History: ---	
Annotation: ---		Inscribed: ---	

Figure 10: Printing the Catalogue Worksheet for an Artifact

When clicking the print button shown in Figure 3, a pop up will appear that will display a pdf preview of the worksheet for an artifact. The worksheet is predetermined, and the website will automatically fill up the fields accordingly.

C.2. Barcode Scanner

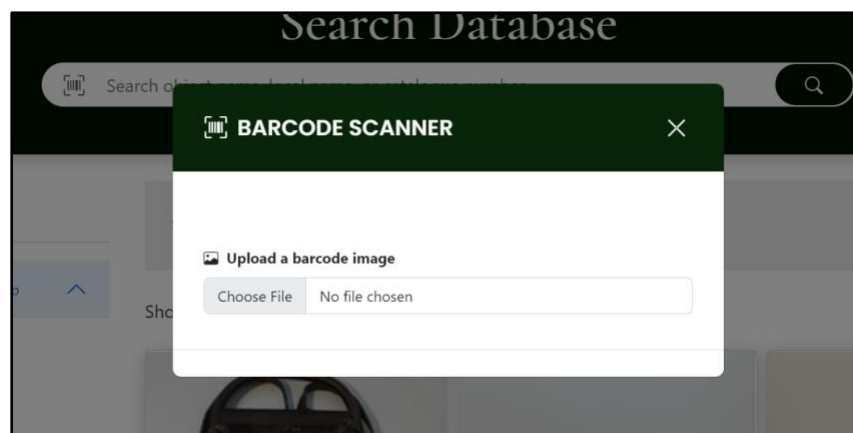


Figure 11: Modal display for barcode scanning

Using an external barcode scanner or any device with the appropriate capability to do so, it is possible to search for its information using it. The main purpose is to speed up the process of artifact identification without constantly relying on image references. It is also possible to upload an image file of the barcode if scanning is not feasible.

C.2. Export to an XLSX File



Figure 12: Export Button Location

A screenshot of an Excel spreadsheet titled 'Purchased Code'. The table contains 22 rows of data, each representing a purchased item. The columns include item ID, date, acquisition status, value, source, object, name, local name, ethnology, category, basketry, exhibition, and current location. The data is as follows:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Purchased	Purchased	Purchased	Purchased	Purchased	Purchased	Purchased	Purchased	Art Source	Object	Nar	Local	Nam	Ethnology	Category	b	Category	b
2	2016-MK-FBAS	001	2016	2016-12-01	Acquired	DV 10078	2000.00		Lyn's Book	Backpack	Chagee	Sagada	Backpack	Basketry				VS1, on ba
3	2016-MK-FBAS	002	2016	2016-12-01	Acquired	DV 10078	600.00		Family Gift	Chicken co	Ifugao	Chicken	Cc	Basketry	Feast of M 2018-02-01	2019-10-01	VS1, on ba	
4	2016-MK-FBAS	003	2016	2016-12-01	Acquired	DV 10078	1800.00		Lyn's Book	Chicken co	Ifugao	Chicken	Cc	Basketry	Feast of M 2018-02-01	2019-10-01	VS1, on ba	
5	2016-MK-FBAS	004	2016	2016-12-01	Acquired	DV 10078	9000.00		Lyn's Book	Ritual back	Takba	Applai Kan	Backpack	Basketry				VS1, on ba
6	2016-MK-FBAS	005	2016	2016-12-01	Acquired				Feather co		Mayaoyao			Basketry				VS1, on ba
7	2016-MK-FBAS	006	2016	2016-12-01	Acquired	DV 10078	1000.00		Family Gift	Basket Cor	Allataw, Kc	Ifugao	Container	Basketry				VS1, on ba
8	2016-MK-FBAS	007	2016	2016-12-01	Acquired	DV 10078	1000.00		Family Gift	Basket Cor	Allataw, Kc	Ifugao	Container	Basketry	Feast of M 2018-02-01	2019-10-01	VS1, on ba	
9	2016-MK-FBAS	008	2016	2016-12-01	Acquired	DV 10078	1500.00		Family Gift	Backpack	Hape-Eng, Bauko, Mt	Backpack	Basketry				VS1, on ba	
10	2016-MK-FBAS	009	2016	2016-12-01	Acquired				Chicken co	Biki, Kubi, Ifugao		Chicken	Cc	Basketry	Feast of M 2018-02-01	2019-10-01	VS1, on ba	
11	2016-MK-FBAS	010	2016	2016-12-01	Acquired	DV 10078	1000.00		Family Gift	Rice storag	Kamuwan, Mt. Provin	Storage	Basketry		Feast of M 2018-02-01	2019-10-01	VS1, on ba	
12	2016-MK-FBAS	011	2016	2016-12-01	Acquired				Rice storag	Kamuwan, Mt. Provin	Storage	(cc)	Basketry		Feast of M 2018-02-01	2019-10-01	VS1, on ba	
13	2016-MK-FBAS	012	2016	2016-12-01	Acquired	DV 10078	1250.00		Family Gift	Fish trap	Gibu, Gubi	Ifugao	Fish Trap	Basketry				VS1, on ba
14	2016-MK-FBAS	013	2016	2016-12-01	Acquired	DV 10078	1250.00		Family Gift	Fish trap	Gibu, Gubi	Ifugao	Fish Trap	Basketry				VS1, on ba
15	2016-MK-FBAS	014	2016	2016-12-01	Acquired				Various su	Winnowin	Bituto, Ligi	Ifugao	Container	Basketry				VS1, on ba
16	2016-MK-FBAS	015	2016	2016-12-01	Acquired	DV 10078	2500.00		Lyn's Book	Rice conta	Kulikug	Applai Kan	Container	Basketry				VS1, on ba
17	2016-MK-FBAS	016	2016	2016-12-01	Acquired	DV 10078	500.00		Purse	Kupit	Bontok	Purse	Basketry					VS1, on ba
18	2016-MK-FBAS	017	2016	2016-12-01	Acquired				Purse	Kupit	Bontok	Purse	Basketry					VS1, on ba
19	2016-MK-FBAS	018	2016	2016-12-01	Acquired	DV 10078	1800.00		Family Gift	Fish trap	Agawin	Applai Kan	Fish Trap	Basketry				VS1, on ba
20	2016-MK-FBAS	019	2016		Acquired				Storage ba	Topil, Tupi	Bontok	Storage	Basketry					VS1, on ba
21	2016-MK-FBAS	020	2016	2016-12-01	Acquired	DV 10078	500.00		Family Gift	Storage ba	Kamuwan, Bontok	Storage	Basketry					VS1, on ba
22	2016-MK-FBAS	021	2016		Acquired				Fish trap	Agawin	Applai Kan	Fish Trap	Basketry					VS1, on ba

Figure 12: XLSX Worksheet for Purchased

The data table for purchased, donated and on-loan can be exported to an xlsx file for archival or printing purposes.